

Understand the Problem

Student work-along checklist

Name: _____

Date started: _____

How to use this checklist

Keep this sheet beside you as you work through the online unit. Tick each box when you have done it.

1. Write first. Answer each question in your own words (online or on paper) **before** you reveal the model answer.

2. Compare. Read the model answer and look for what is the same and what is different.

3. Annotate. Add one thing you missed or would change. A different answer is not always a wrong answer.

Before you start: read the Introduction and learning objectives, and the Recap of Unit 2 if you need a reminder of the workflow.

The whole unit takes about **3 hours**. You do not have to do it in one sitting. Use the section headings to find your place again.

Keep in view: the Project Brief (summary)

A **health research organisation** has collected health and lifestyle data from many participants over several years.

They want to know whether this data can answer questions about the **factors connected to people's overall health**, and provide evidence for future health and wellbeing initiatives.

They have asked your team to **review the project, identify any missing information, and advise** how the investigation should proceed.

The brief is deliberately vague. Deciding what extra information is needed is part of your job.

Dataset: health_data.csv, about 10,000 rows (one per participant) and 64 columns.

1 Understand the Problem

Activity 3.1: explore the data in a spreadsheet

- ☐ Make a folder for this course. Download **health_data.csv** and save a clean copy in it. Do not edit this copy.
- ☐ Open it in Google Sheets or Excel and scroll through the columns.
- ☐ Read the note on snake_case column names (Health_Score, not Health Score).

Activity 3.1 questions ☐ 3.1(a) ☐ 3.1(b) ☐ 3.1(c) ☐ 3.1(d) ☐ 3.1(e) ☐ 3.1(f)

Activity 3.2: load the data into Orange

If you cannot install or open Orange, tell your teacher or mentor and carry on with the rest of the unit. Activity 3.2 repeats in Orange much of what you found in the spreadsheet in Activity 3.1.

Orange steps ☐ Part 1 ☐ Part 2 ☐ Part 3 ☐ Part 4

Part 4: save the .ows file in your course folder. You will need it in Unit 4.

Activity 3.2 questions ☐ 3.2(a) ☐ 3.2(b) ☐ 3.2(c) ☐ 3.2(d) ☐ 3.2(e) ☐ 3.2(f)

2 Analyse the Problem

- ☐ I can identify the four real-world constraints: **time, cost, access, privacy and ethics**.

Activity 3.3 questions ☐ 3.3(a) ☐ 3.3(b) ☐ 3.3(c) ☐ 3.3(d) ☐ 3.3(e) ☐ 3.3(f) ☐ 3.3(g)

3 Can building a machine learning model add value?

- ☐ I can name the three kinds of question: **what the data already shows**, **questions with a known rule** (for example, BMI), and **estimating an unknown value**, which is where machine learning can help.
- ☐ I can explain why deciding **not** to use machine learning is a valid outcome.

4 Understanding Data Stakeholders

- ☐ The four types. For each one, I can say who they are and what they need: **data subjects, data producers, data agents, data users**.

Activity 3.4 questions ☐ 3.4(a) ☐ 3.4(b) ☐ 3.4(c) ☐ 3.4(d) ☐ 3.4(e)

5 Defining Success and Managing Risk

- ☐ I can say what makes good success criteria: **clear, realistic, relevant, measurable**.

Activity 3.5 questions ☐ 3.5(a) ☐ 3.5(b) ☐ 3.5(c) ☐ 3.5(d) ☐ 3.5(e)

Before you check 3.5(b): is each of your criteria measurable? Could someone tell whether it had been met?

6 Formulating the Research Question

Activity 3.6: check your research question

After you write 3.6(a), check your draft before you reveal the model answer:

- ☐ **Clear:** someone who is not a data scientist would understand it.
- ☐ **Focused:** it asks about one main thing, not several.
- ☐ **Answerable:** the data it needs is in health_data.csv.
- ☐ **Relevant:** it links to what the organisation said it wants to know.
- ☐ It names **what is being estimated or explained** (for example, a column in the dataset).
- ☐ It does **not assume the answer**. For example, "How much does exercise improve..." already assumes that exercise helps.

If any box is unticked, redraft your question before you check the model answer.

Activity 3.6 questions ☐ 3.6(a) ☐ 3.6(b) ☐ 3.6(c) ☐ 3.6(d)

7 Writing the Problem Statement

Activity 3.7: writing frame

Draft your problem statement one part at a time, then join the parts into one or two sentences for 3.7(a).

Part	Your sentence (use the starter if it helps)
1. The research question What is being investigated?	<i>Investigate whether ...</i>
2. The data What data will be used?	<i>The investigation will use ...</i>
3. The purpose What will the investigation find out?	<i>... to determine whether ...</i>
4. The intended outcome Who will it help, and how?	<i>... that support ...</i>

☐ My statement says “investigate whether”, not “show that”. It does not promise a result.

Activity 3.7 questions ☐ 3.7(a) ☐ 3.7(b) ☐ 3.7(c)

☐ I have compared my research question and problem statement with the agreed versions at the end of the section.

8 Unit 3 Review

☐ I have answered the four **Reflection Questions** (they are good to discuss with a partner or mentor).

How confident am I?

Tick each one you are confident about. Leave a box empty if you are not sure yet, and ask your teacher or mentor about it.

- | | |
|--|--|
| ☐ I can explain why understanding the problem comes first | ☐ I can define success criteria |
| ☐ I can analyse the requirements of a project | ☐ I can write a focused, answerable research question |
| ☐ I can identify stakeholder types and their roles | ☐ I can write a clear problem statement |
| ☐ I can decide what data is needed | ☐ I can identify risks and apply “Do No Harm” |
| ☐ I can recognise practical constraints | |

Ready for Unit 4?

- ☐ A clean copy of **health_data.csv** is in my course folder.
- ☐ My Orange **.ows** file is saved in the same folder.
- ☐ I know where to find the **agreed research question and problem statement**.
- ☐ I have noted anything I am still unsure about to ask my teacher or mentor.